Status	Finished
Started	Saturday, 10 October 2026, 1:53 PM
Completed	Saturday, 10 October 2026, 2:00 PM
Duration	6 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(1)
- c. O(n log n)
- d. O🥐

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By using a hash function
- b. By comparing each element sequentially
- c. By dividing the list length by two ✓
- d. By starting from the first element

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n ✓
- b. log n
- c. 1
- d. n/2

Question **4**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 1000
d. 10 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. The list must be sorted
c. Number values should only be present

Number values should only be present

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is somewhere in the middle of the array ✓
c. When the item is not the array at all
d. When the item is the last element in the array

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the list is sorted
c. When the search algorithm finishes
d. When the element is found in the list ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Linear search ✓
- b. Quick search
- c. Bubble search
- d. Merge search

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search stops
- b. The list is sorted
- c. The search continues in the left or right sublist ✓
- d. The search continues from the beginning

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list is very small
- c. When the list contains duplicate elements
- d. When the list is unsorted

Question **11**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is small and sorted
- c. When the list is large and unsorted
- d. When the list is large and sorted

Question **12**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 89 and 99

c. 90 and 99 ✓

d. 89 and 94

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers

b. It is slower than linear search for small lists

c. It requires the list to be sorted ✓

d. It can only be applied to large lists

Question **14**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining

b. Searching ✓

c. Discovering

d. Finding

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search

b. Linear search ✓

c. Binary search

d. Hashing